PROJECT DESIGN PHASE

Script-Controlled ACL – Restrict Record Access Based on Field Value

1. Design Overview

The Project Design Phase defines the structure and working mechanism of the Script-Controlled ACL system. The design focuses on controlling access to records by checking specific field values and user conditions through a custom ACL script. The system grants access only when the defined conditions are satisfied and the ACL script evaluates to true.

2. System Architecture

The system follows an access-control based architecture consisting of the User Interface, Authentication Module, Record Management Module, ACL Configuration Module, ACL Script Evaluation Module, and Database.

The user first authenticates with the system. When the user requests a protected record, the system identifies the user and retrieves the required record information. The ACL script then checks the configured field values and user conditions. Based on the result, the system either grants or denies access.

3. Module Design

• User Authentication Module: Authenticates users and identifies the current user.
• Record Management Module: Stores and manages application records and their field values.
• ACL Configuration Module: Allows authorized administrators to define access-control conditions.
• Script Evaluation Module: Executes the custom ACL script and evaluates the required conditions.
• Access Decision Module: Grants access when the script returns true and denies access when it returns false.
• Audit/Logging Module: Records important access decisions and related activities for monitoring.

4. Access Control Flow

1. The user logs into the system.
2. The user requests access to a protected record.
3. The system identifies the current user.
4. The system retrieves the required record and field values.
5. The ACL script checks the specified field values and user conditions.
6. The script evaluates the access conditions.
7. If the result is true, access to the record is granted.
8. If the result is false, access to the record is denied.

5. Data Design

The system maintains information required for authentication, users, records, field values, ACL rules, and access activities. User information is used to identify the requester, while record information contains the fields used by the ACL conditions. ACL configuration stores the rules required for evaluating access requests.

6. User Interface Design

The interface should provide a login screen for authentication and suitable pages for viewing records. Authorized administrators should have an interface for configuring and managing ACL conditions. When access is denied, the system should display a clear authorization message without exposing restricted record information.

7. Security Design

Security is implemented by validating access requests through the ACL before allowing protected records to be accessed. The ACL script checks both record field values and user conditions. Direct access to restricted records must not be permitted when the script evaluates to false. Administrative functions should also be limited to authorized users.

8. Expected Design Outcome

The proposed design provides a structured and secure mechanism for record-level access control. By combining user validation, field-value checking, and custom ACL scripts, the system can provide flexible access decisions. The design can also be extended with additional conditions and rules as future requirements increase.